CAPACITANCE

Capacitor and Capacitance

A capacitor is a device used to store electric charge and electrical energy. It consists of two conductors separated by an insulating medium. The capacitance of a capacitor is the ratio of charge stored on either conductor to the potential difference between them.

$$C = Q / V$$

SI Unit of Capacitance

The SI unit of capacitance is farad (F). A capacitor has capacitance of one farad when a charge of one coulomb produces a potential difference of one volt between its plates.

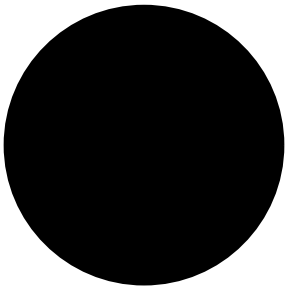
Capacitance of Isolated Spherical Conductor

For an isolated spherical conductor of radius R in a medium of permittivity epsilon, the capacitance depends on its radius and the permittivity of the surrounding medium. For an isolated sphere in vacuum, the capacitance is proportional to its radius.

$$C = 4 \pi \varepsilon_0 R$$

Isolated Spherical Conductor

Isolated Spherical Conductor



Parallel Plate Capacitor

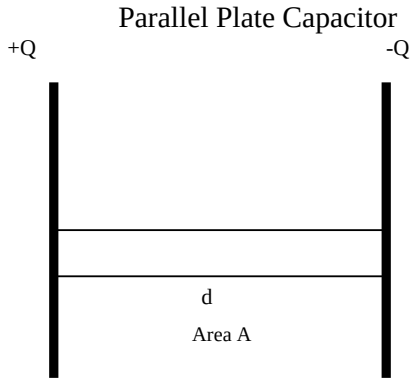
A parallel plate capacitor consists of two large conducting plates separated by a small distance d. When the separation is small compared with the dimensions of the plates, the electric field between the plates is approximately uniform.

$$C = \varepsilon_0 A / d$$

Factors Affecting Capacitance

The capacitance of a parallel plate capacitor increases with the area A of the plates and decreases with the separation d between them. It also depends on the permittivity of the medium between the plates.

Parallel Plate Capacitor



Effect of Dielectric

When a dielectric material is completely inserted between the plates of a capacitor, the capacitance increases by the dielectric constant K of the material.

$$C = K \varepsilon_0 A / d$$

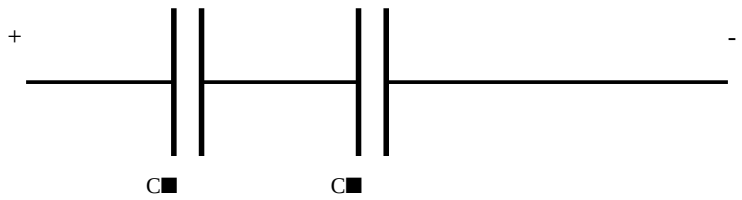
Combination of Capacitors — Series

When capacitors are connected in series, the same charge is stored on each capacitor. The total potential difference is the sum of the potential differences across individual capacitors.

$$1 / C_{eq} = 1 / C_1 + 1 / C_2 + 1 / C_3$$

Capacitors in Series

Capacitors in Series



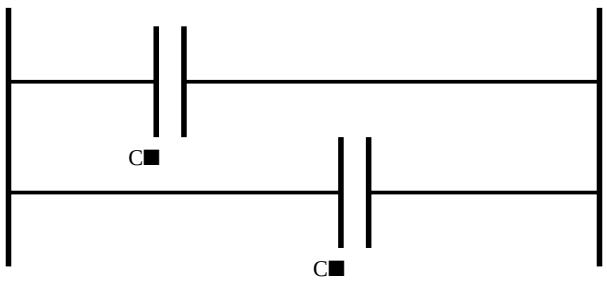
Combination of Capacitors — Parallel

When capacitors are connected in parallel, the potential difference across each capacitor is the same. The total charge is the sum of the charges stored on individual capacitors.

$$C_{eq} = C_1 + C_2 + C_3$$

Capacitors in Parallel

Capacitors in Parallel



Energy Stored in a Capacitor

A charged capacitor stores electrical potential energy. The energy stored depends on the capacitance and the charge or potential difference.

$$U = 1 / 2 \, C \, V^2$$

$$U = 1 / 2 \, Q \, V$$

$$U = Q^2 / 2 \, C$$

Energy Density

Energy density is the electrical energy stored per unit volume of the electric field region between the capacitor plates.

$$u = 1 / 2 \, \varepsilon_0 \, E^2$$

Charging and Discharging — Basic Concept

During charging, a capacitor stores charge and its potential difference increases until it reaches the applied voltage. During discharging, the stored charge and potential difference decrease through the connected circuit.

Important Formulae

Key formulas for Capacitance are listed below.

$$C = Q / V$$

$$C = 4 \pi \varepsilon_0 \, R$$

$$C = \varepsilon_0 A / d$$

$$C = K \varepsilon_0 A / d$$

$$1 / C_{eq} = 1 / C_1 + 1 / C_2 + 1 / C_3$$

$$C_{eq} = C_1 + C_2 + C_3$$

$$U = 1 / 2 C V^2$$

$$U = 1 / 2 Q V$$

$$U = Q^2 / 2 C$$

$$u = 1 / 2 \varepsilon_0 E^2$$

Quick Revision Points

Capacitance is charge per unit potential difference. Parallel plate capacitance increases with plate area and decreases with plate separation. A dielectric increases capacitance. In series combination, charge is the same on each capacitor. In parallel combination, potential difference is the same across each capacitor. A capacitor stores electrical energy.

MHT-CET Important Points / Numerical Focus

Focus on capacitance formulas, dielectric constant, equivalent capacitance in series and parallel combinations, charge and potential difference, energy stored in a capacitor and energy density. Pay attention to units, circuit connections and changes in capacitance when the geometry or dielectric medium is changed.